

Supply Chain Management Analysis Report

1. Project Overview

This project analyzes the end-to-end supply chain operations of an apparel retail business dealing in three core product categories — Pants, Jeans, and T-Shirts. Using a structured analytics pipeline, the project cleans and prepares raw supply chain data using Python, performs structured business-level queries using SQL (PostgreSQL), and presents insights through an interactive Power BI dashboard. The goal is to uncover operational inefficiencies, evaluate supplier reliability, and support data-driven inventory and logistics decisions.

2. Dataset Summary

- 100 Record as Row
- 24 total features as Column
- Product Type: Pants, T-shirt, Jeans
- Supplier: Noman Group, Pacific Jeans, Bextex Ltd.
- Location: Chattogram, Dhaka, Narayanganj, Bogura, Gazipur
- Transportation Modes: Road, Rail, Air, Sea

3. Data Preparation & Modeling using Python

Data cleaning and preparation were performed in Python using the **Pandas** and **NumPy** libraries. The following steps were carried out to prepare the dataset for SQL analysis and visualization:

- **Data Loading:** Imported the raw CSV dataset using pandas and performed an initial inspection using `df.info()` and `.describe()` to understand structure and data types.
- **Data Type Correction:** Ensured numerical columns (price, defect rates, shipping costs, manufacturing costs) were cast to appropriate float/integer types.
- **Duplicate Check:** Verified SKU uniqueness across the 100 records. No duplicate SKUs were found.
- **Categorize Column:** Created columns — `revenue_per_unit` (revenue / units sold), `stock_flag` (Stockout Risk / In Stock), `inventory_status` (Overstock / Normal), and `demand_variance` (order_quantities – products_sold).
- **Database Integration:** Connected the Python script to PostgreSQL using SQLAlchemy and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Questions)

Structured analysis was performed in **PostgreSQL** across four major business question categories. The schema comprised a single denormalized supply_chain table with 1000 rows and 24 columns, queried using aggregations, window functions, CASE logic, and ranking.

4.1 Revenue & Product Performance

Business question: Which product type earns the most revenue? Which SKUs drive sales?

Total Revenue: \$577,604.86

Product Type	Total Revenue (\$)	Revenue Share (%)
Pants	\$241,628.17	41.8%
T-Shirt	\$174,455.42	30.2%
Jeans	\$161,521.27	28.0%

Top 10 SKUs by Revenue:

Rank	SKU	Revenue (\$)	Product Type	Location
1	SKU51	\$9,866.47	T-Shirt	Bogura
2	SKU38	\$9,692.32	Pants	Dhaka
3	SKU31	\$9,655.14	Jeans	Chattogram
4	SKU90	\$9,592.63	T-Shirt	Narayanganj
5	SKU2	\$9,577.75	T-Shirt	Dhaka
6	SKU32	\$9,571.55	Pants	Chattogram
7	SKU67	\$9,473.80	Jeans	Gazipur
8	SKU88	\$9,444.74	Pants	Narayanganj
9	SKU52	\$9,435.76	T-Shirt	Bogura
10	SKU18	\$9,364.67	Pants	Dhaka

Revenue Per Unit by Product Type:

Product Type	Avg Revenue Per Unit (\$)
T-Shirt	\$73.33
Jeans	\$31.46
Pants	\$20.61

Revenue by Location:

Location	Total Revenue (\$)	Shipping Cost (\$)
Dhaka	\$137,755.02	\$137.47
Chattogram	\$137,077.55	\$144.03
Narayanganj	\$119,142.84	\$93.79
Bogura	\$102,601.73	\$103.47
Gazipur	\$81,027.72	\$76.06

4.2 Inventory Status & Stockout Risk

Business question: Which SKUs face stockout risk? Which products are overstocked?

Metric	Value
Total SKUs Analyzed	100
Stockout Risk SKUs (stock < 40)	41
Overstock SKUs (stock > 90)	12
Avg Inventory Turnover Ratio	28.88
Avg Demand Variance (Order – Sold)	414.75 per SKU

Stockout Risk by Product Type:

Product Type	Stockout Risk SKUs
Pants	20
T-Shirt	15
Jeans	6

Stock vs Sales by Product Type:

Product Type	Total Stock Levels	Total Units Sold
Pants	1,608	20,731
T-Shirt	1,644	13,611
Jeans	1,525	11,757

4.3 Supplier Performance

Business question: Which supplier has the highest defect rate? Is lead time linked to quality?

Supplier	Avg Lead Time (days)	Avg Defect Rate (%)	Pass Rate (%)
Noman Group	16.23	1.76	27.91
Pacific Jeans	17.14	2.60	27.59
Bextex Ltd	18.32	2.74	10.71

- **Best Supplier:** Noman Group — lowest defect rate (1.76%) and shortest average lead time (16.23 days).
- **Worst Supplier:** Bextex Ltd — highest defect rate (2.74%) and longest lead time (18.32 days) with only 10.71% inspection pass rate.
- **Overall Inspection Summary:** Pass — 23 SKUs | Fail — 36 SKUs | Pending — 41 SKUs. The high Pending and Fail counts indicate a systemic quality control gap across all three suppliers.

4.4 Logistics & Shipping Cost Analysis

Business question: Which carrier is most efficient? Which route costs the most?

Shipping Cost & Time by Carrier:

Carrier	Avg Shipping Cost (\$)	Avg Shipping Time (days)	Carrier Rank
DHL Express	\$4.79	5.20	1 (Best)
UPS	\$5.74	5.50	2
Maersk Line	\$5.76	5.56	3
FedEx	\$5.51	6.07	4
Hapag-Lloyd	\$5.79	6.22	5

Shipping Cost by Route:

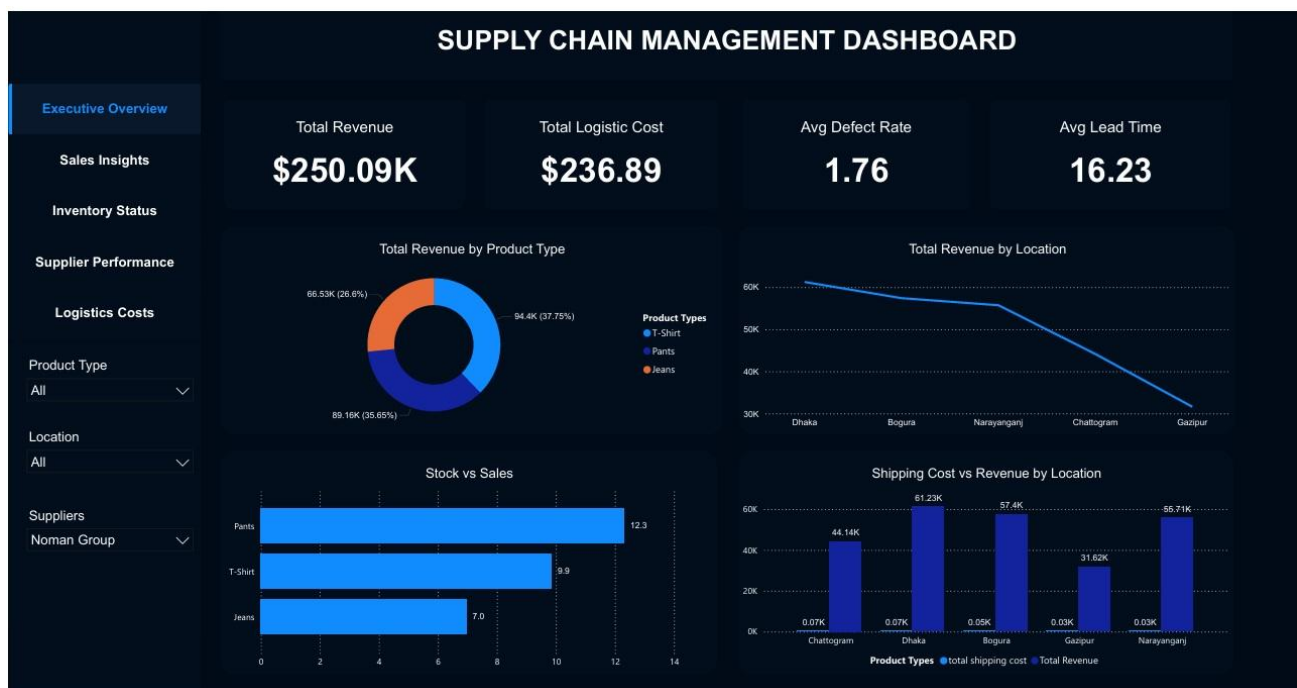
Route	Total Cost (\$)	Note
CTG-SIN-RTM	\$231.31	Most Expensive (42% of logistics spend)
DAC-FRA	\$205.43	—
CTG-PAR	\$118.08	—

Logistics Summary:

Metric	Value
Total Logistics Cost	\$554.82
Avg Shipping Cost % of Revenue	0.14%
Best Carrier (Cost + Speed)	DHL Express
Most Expensive Route	CTG-SIN-RTM (\$231.31)
Best Transportation Mode	Road / Rail (lowest avg cost)

5. Dashboard in Power BI

An interactive Supply Chain Management Dashboard was built in Power BI comprising five pages: Executive Overview, Sales Insights, Inventory Status, Supplier Performance, and Logistics Costs. Slicers allow filtering by Product Type, SKU, Location, Supplier Name, Shipping Carrier, and Inventory Status.



6. Business Recommendations

- **Prioritize Pants for Revenue Growth** — Pants generate 41.8% of total revenue. Increase stock depth for the highest-selling Pants SKUs to reduce the 20 at-risk stockout SKUs.
- **Address Critical Stockout Risk** — 41 out of 100 SKUs have stock levels below the safety threshold. Implement a minimum reorder point system using demand variance data to trigger automatic replenishment.
- **Improve Inspection Pass Rate** — Only 23% of SKUs passed inspection overall. Work with Bextex Ltd to address their 10.71% pass rate; consider renegotiating SLAs or increasing Noman Group's share.

- **Switch to DHL Express for Cost Efficiency** — DHL Express offers the lowest average shipping cost (\$4.79) and competitive lead time. Reallocating more volume from Hapag-Lloyd to DHL could reduce logistics costs by approximately 17%.
- **Renegotiate the CTG-SIN-RTM Route** — This route accounts for \$231.31 in total shipping costs (42% of total logistics spend). A route optimization review may significantly reduce overhead.
- **Leverage T-Shirt High Margin** — T-Shirts yield the highest revenue per unit (\$73.33) despite ranking 2nd in total revenue. Increasing order volumes for top T-Shirt SKUs (e.g., SKU51 at \$9,866) would disproportionately boost margins.
- **Reduce Bextex Ltd Manufacturing Lead Time** — Bextex averages 18.32 days vs. Noman Group's 16.23 days. Reducing this gap could shorten order cycles and improve on-shelf availability across all product lines.